

**Question 1:**

Define what it means for a function a multi-variate function  $f$  to be symmetric.

**Question 2:**

Show that for any symmetric polynomial, say  $p$ , we can write  $p$  in the following form

$$p(x_1, \dots, x_n) = a_1 V_1 + \dots a_n V_n$$

Where  $V_k = \sum_A \prod_j^k x_j$ , i.e. the sum of all monomials of degree  $k$  (their permutation)

**Question 3:**

Define inclusion map

**Question 4:**

We claim that the map  $\psi : \mathbb{Z} \rightarrow R$  is unique and given by  $\psi : n \mapsto \sum^n 1_R \in R$ .

Show why the map  $\Phi : \mathbb{Z} \rightarrow R$  given by  $\Phi : n \mapsto 0_R^n \in R$ , does satisfy.

**Question 5:**

In exact case, what is it that we want?